

Homework #2
Math 104: Intro to Analysis

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1 Problem 1

Let S be the set of all *binary sequences*, that is $S = \{(x_1, x_2, x_3, \dots) : x_j \in \{0, 1\} \text{ for all } j\}$.

- (a) Using Cantor's diagonalization argument, show that S is uncountable.
- (b) Let $T \subset S$ be the set of all binary sequences with *finite support*, that is let $T = \{(x_1, x_2, x_3, \dots) \in S : \text{only finitely many of the } x_j \text{ are nonzero}\}$. Show that T is countable.
Hint: Let $S_n = \{(x_1, x_2, x_3, \dots) \in S : x_j = 0 \text{ for all } j \geq n+1\}$.

1.1 Solution (a)

Alvin's Proposition: Define $r : \{0, 1\} \rightarrow \{0, 1\}$ where $r(n) = \begin{cases} 0 & \text{if } n = 1 \\ 1 & \text{if } n = 0 \end{cases}$. For any $x \in \{0, 1\}$, $x \neq r(x)$.

Proof by Cases. We consider the cases of $x = 0$ and $x = 1$.

Case 1: $x = 0$. If $x = 0$, then $r(x) = r(0) = 1$. Therefore, $x \neq r(x)$.

Case 2: $x = 1$. If $x = 1$, then $r(x) = r(1) = 0$. Therefore, $x \neq r(x)$.

Ergo, for any $x \in \{0, 1\}$, $x \neq r(x)$.

OEΔ

This proof is done through Cantor's Diagonalization Argument, which is a proof by contradiction.

Proof by Contradiction. Suppose for contradiction that S is countable. That means that we can enumerate the members of S . We can tabulate these.

$$\begin{array}{llllll} S_1 = & (x_{11}, & x_{12}, & x_{13}, & x_{14}, & x_{15}, & \dots) \\ S_2 = & (x_{21}, & x_{22}, & x_{23}, & x_{24}, & x_{25}, & \dots) \\ S_3 = & (x_{31}, & x_{32}, & x_{33}, & x_{34}, & x_{35}, & \dots) \\ S_4 = & (x_{41}, & x_{42}, & x_{43}, & x_{44}, & x_{45}, & \dots) \\ S_5 = & (x_{51}, & x_{52}, & x_{53}, & x_{54}, & x_{55}, & \dots) \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{array}$$

Note the binary sequence elements which can be written in our enumeration as x_{jj} for some natural number j .

$$\begin{array}{llllll} S_1 = & (\textcircled{x_{11}}, & x_{12}, & x_{13}, & x_{14}, & x_{15}, & \dots) \\ S_2 = & (x_{21}, & \textcircled{x_{22}}, & x_{23}, & x_{24}, & x_{25}, & \dots) \\ S_3 = & (x_{31}, & x_{32}, & \textcircled{x_{33}}, & x_{34}, & x_{35}, & \dots) \\ S_4 = & (x_{41}, & x_{42}, & x_{43}, & \textcircled{x_{44}}, & x_{45}, & \dots) \\ S_5 = & (x_{51}, & x_{52}, & x_{53}, & x_{54}, & \textcircled{x_{55}}, & \dots) \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{array}$$

Let's create a new binary sequence S_α . Define $r : \{0, 1\} \rightarrow \{0, 1\}$ where $r(n) = \begin{cases} 0 & \text{if } n = 1 \\ 1 & \text{if } n = 0 \end{cases}$.

By Alvin's Proposition, for any $x \in \{0, 1\}$, $x \neq r(x)$. Take each element written in our enumeration as x_{jj} for some natural number j . Define each member $x_{\alpha j}$ of S as $r(x_{jj})$.

$$S_\alpha = (x_{\alpha 1}, x_{\alpha 2}, x_{\alpha 3}, x_{\alpha 4}, x_{\alpha 5}, \dots) = (r(x_{11}), r(x_{22}), r(x_{33}), r(x_{44}), r(x_{55}), \dots) \quad (1)$$

Since for any $x \in \{0, 1\}$, $x \neq r(x)$, that means that for any natural number k , $x_{kk} \neq x_{\alpha k}$. Therefore, for any binary sequence S_k , there exists some element x_{kk} that is different from its respective element $x_{\alpha k}$ of S_α . That means that S_α is a member of S that differs from each S_k by at last one digit. That means that S_α cannot occur in our enumeration. This contradicts that S is the set of all binary sequences. Therefore, our initial assumption is false and S is uncountable. OEΔ

1.2 Solution (b)

Proof. Recall that the rational numbers are countable. This means that if we can map the binary sequences onto the rational numbers, we can prove that they are countable.

Since only finitely many of x_j are nonzero, there is a point $n > 0$ beyond which for all $j > n$, $x_j = 0$. Take any element t in T such that $t = (x_1, x_2, \dots)$. We can define a number $N < 1$, turning every x_j into a digit. Since for all $j > n$, $x_j = 0$, there is a terminal point in N beyond which every subsequent digit would be 0.

$$N = 0.x_1x_2x_3 \dots x_{n-1}x_n0000 \dots \quad (2)$$

We can multiply both sides of N by 10^n . This allows us to shift the decimal point to just after x_n .

$$N \times 10^n = 10^n \times 0.x_1x_2x_3 \dots x_{n-1}x_n0000 \dots \quad (3)$$

$$= x_1x_2x_3 \dots x_{n-1}x_n.0000 \dots \quad (4)$$

Since it has nothing but 0 past the decimal point, that means that $N \times 10^n$ is a natural number. Furthermore, since $n > 0$, 10^n is a natural number. We can divide both sides by 10^n .

$$\frac{N \times 10^n}{10^n} = \frac{x_1x_2x_3 \dots x_{n-1}x_n.0000 \dots}{10^n} \quad (5)$$

$$N = \frac{x_1x_2x_3 \dots x_{n-1}x_n}{10^n} \quad (6)$$

Therefore, N is an integer over a natural number, meaning that it is a rational number. In order to establish that this is an injection, we must establish that for s and t in T where $s \neq t$, then $N_s \neq N_t$. If $s \neq t$, then there is a smallest number k where $s_k \neq t_k$. That leads to digit k of N_s not being equal to digit k of N_t . This means that $N_s \neq N_t$.

This establishes an injection from the binary sequences into the rational numbers. Therefore, because the rational numbers are countable, T is countable. OEΔ

2 Problem 2

Let (X, d) be a metric space. For a subset $E \subset X$, define the *closure* of E , denoted $\overline{E}$, to be $E \cup E'$.

- (a) Show that $\overline{E}$ is closed, and that $\overline{E} = E$ if and only if E is closed.
- (b) Show that $\overline{E}$ is the *smallest* closed set containing E in the following sense: if F is closed and $E \subset F$, then $\overline{E} \subset F$.
- (c) Show that $\overline{E}$ is the intersection of all closed subsets of X containing E .
- (d) Show that for any $E, F \subset X$, we have $\overline{E \cup F} = \overline{E} \cup \overline{F}$ (this is much easier if you use part (b)).
- (e) Define the *closed ball of radius r centered at x* to be $D_r(x) = \{y \in X : d(y, x) \leq r\}$. Show that $D_r(x)$ is closed and that $\overline{B_r(x)} \subset D_r(x)$ (using part (b) is again much simpler here).

2.1 Solution (a)

First we prove that $\overline{E}$ is closed.

Proof. We do this by proving that the complement of $\overline{E}$ is open. Take any g in $\overline{E}^c$. Since $\overline{E}$ is equal to the union of E and the set of all its limit points, g is neither a limit point of E nor in E . Not all $r > 0$ create an open ball $B_r(g)$ that has an intersection with $\overline{E}$. That means that there exists some $r > 0$ such that the open ball $B_r(g)$ is a subset of $\overline{E}^c$. This means that $\overline{E}^c$ is open. Because a set is open if and only if its complement is closed, $\overline{E}$ is closed. OEΔ

Next, we prove that $\overline{E} = E$ if E is closed.

Proof. If E is closed, that means that it contains all its limit points. Recall that $\overline{E} = E \cup E'$, with E' referring to the set of all limit points of E . This means that E' is a subset of E . Therefore, $E \cup E'$ is equal to E . Ergo, transistively, $\overline{E} = E$. OEΔ

Third, we prove that E is closed if $\overline{E} = E$.

Proof. If $\overline{E} = E$, then E is equal to the union of E and the set of all its limit points E' .

Take any point p in E' . Since p is in E' , that means p is in the union of E and E' . Since E is equal to the union of E and E' , that means that p is in E . This means that if a point is a limit point of E , then it is in E . That means that E contains all of its limit points. Ergo, E is closed. OEΔ

Last, unification.

Proof. We know that $\overline{E} = E$ if E is closed. We know that E is closed if $\overline{E} = E$. Ergo, $\overline{E} = E$ if and only if E is closed. OEΔ

2.2 Solution (b)

Proof. We will prove that if F is closed and E is a subset of F , then $\overline{E}$ is a subset of F .

Observe the limit points of E (denoted E'), whether they exist or not. Take any point p in E' . Since p is a limit point of E , that means that for every $r > 0$, there exists an element q in E that is not equal to p such that the open ball $B_r(p)$ contains q . Since E is a subset of F , this means that if q is in E , then q is also in F . Therefore, for every $r > 0$, there exists an element q in F that is not equal to p such that the open ball $B_r(p)$ contains q . This means that p is a limit point of F , and therefore that every limit point of E is a limit point of F . Since F is closed, that means it contains all its limit points. Therefore E' is a subset of F . Since E and E' are both subsets of F , that means their union is also a subset of F . Since $\overline{E}$ is the union of E and E' , that means $\overline{E}$ is a subset of F . OEΔ

2.3 Solution (c)

Proof. In part (b), we proved that if F is closed and E is a subset of F , then $\overline{E}$ is a subset of F . We also know that the intersection of any number of closed sets is also a closed set. Suppose that we give the set of all closed subsets of X containing E the name G .

The only thing we know all the sets in G have in common is that they contain E as a subset. Therefore, we can say that the intersection of all elements of G contains E as a subset. As proven in part (b), if a closed set has E as a subset, then $\overline{E}$ is a subset of the closed set. Therefore, $\overline{E}$ is a subset of the intersection of all elements of G .

Since G is the collection of all closed subsets of X and we have established that $\overline{E}$ is both closed and a subset of X , we know that $\overline{E}$ is an element of G . By that, we know that if an element p of X is not in $\overline{E}$, it is not in the intersection of all elements of G . Therefore, the intersection of all elements of G is a subset of $\overline{E}$.

Since the intersection of all elements of G is both a subset and superset of $\overline{E}$, that means the intersection of all elements of G is equal to $\overline{E}$. OEΔ

2.4 Solution (d)

Baron's Proposition: $A \subset C$ and $B \subset C$ if and only if $A \cup B \subset C$.

Proof. First the forward direction. Suppose that $A \subset C$ and $B \subset C$. Any element in $A \cup B$ is either in A or in B . If it is in A , since $A \subset C$, it is also in C . If it is in B , since $B \subset C$, it is also in C . Either way, it is in C , so every element in $A \cup B$ is in C . Therefore, $A \cup B$ is a subset of C .

Next the backwards direction. Suppose that $A \cup B \subset C$. Every element in A is in $A \cup B$. Since $A \cup B \subset C$, every element in A is in C . Similarly, every element in B is in $A \cup B$. Since $A \cup B \subset C$, every element in B is in C . Therefore, $A \subset C$ and $B \subset C$.

Ergo, $A \subset C$ and $B \subset C$ if and only if $A \cup B \subset C$. OEΔ

First, we demonstrate that $\overline{E \cup F}$ is a subset of $\overline{E} \cup \overline{F}$.

Proof. Since the union of finite closed sets is also a closed set, $\overline{E \cup F}$ is a closed set. Since $\overline{E}$ is equal to the union of E and E' , then E is a subset of $\overline{E}$. This means that E is a subset of $\overline{E \cup F}$. By a similar argument, F is a subset of $\overline{E \cup F}$. Since E and F are both subsets of $\overline{E \cup F}$, Baron's Proposition tells us that $E \cup F$ is a subset of $\overline{E \cup F}$. Since $\overline{E \cup F}$ is a closed set, according to part (b), $\overline{E \cup F}$ is also a subset of $\overline{E \cup F}$. OEΔ

Next, we demonstrate that $\overline{E} \cup \overline{F}$ is a subset of $\overline{E \cup F}$.

Proof. First, we can write out $\overline{E \cup F}$ in long form.

$$\overline{E \cup F} = (E \cup F) \cup (E \cup F)' \quad (7)$$

Since it is written as a union including E , we can say that E is a subset of $\overline{E \cup F}$. Since $\overline{E \cup F}$ is closed, by part (b), $\overline{E}$ is a subset of $\overline{E \cup F}$. By a similar argument, $\overline{F}$ is a subset of $\overline{E \cup F}$.

By Baron's Proposition, that means that $\overline{E \cup F}$ is a subset of $\overline{E \cup F}$. OEΔ

Last, we combine the two subset statements.

Proof. We know that $\overline{E \cup F}$ is a subset of $\overline{E \cup F}$ and that $\overline{E \cup F}$ is a subset of $\overline{E \cup F}$. Since each is a subset the other, that means that $\overline{E \cup F} = \overline{E \cup F}$. OEΔ

2.5 Solution (e)

First we prove that $D_r(x)$ is closed.

Proof. For $D_r(x)$ to be closed, its complement would have to be open. We can define its complement.

$$D_r(x)^c = \{y \in X : d(y, x) > r\} \quad (8)$$

Take any point p in $D_r(x)^c$. We know that $d(p, x) > r$. That means that $d(p, x) - r > 0$. Suppose that we create an open ball of radius $s = d(p, x) - r$.

$$B_s(p) = \{y \in X : d(y, p) < d(p, x) - r\} \quad (9)$$

If we can prove that $d(y, x) > r$ for any $y \in B_s(p)$, we can prove that the open ball is a subset of $D_r(x)^c$. By the triangle inequality, with x as our third point, $d(x, p) \leq d(y, x) + d(y, p)$. This means that $d(y, x) \geq d(x, p) - d(y, p)$. Since $d(p, x) > r$, we know $d(y, x) > d(x, p) - d(x, p) + r = r$.

Therefore, every point in $B_s(p)$ is in $D_r(x)^c$. This means that $D_r(x)^c$ is open. Ergo, $D_r(x)$ is closed. OEΔ

We can now prove that $\overline{B_r(x)}$ is a subset of $D_r(x)$.

Proof. First, we prove that $B_r(x)$ is a subset of $D_r(x)$. Take any point y in $B_r(x)$. We can say that $d(x, y) < r$. That means that $d(x, y) \leq r$. Therefore, y is in $D_r(x)$. This means that $B_r(x)$ is a subset of $D_r(x)$.

Part (b) tells us that if F is closed and E is a subset of F , then $\overline{E}$ is a subset of F . $D_r(x)$ is closed. $B_r(x)$ is a subset of $D_r(x)$. Therefore, $\overline{B_r(x)}$ is a subset of $D_r(x)$. OEΔ

3 Problem 3

Let S be the set of all four letter strings of letters (which we will call *words*). Thus $S = \{\text{abcd}, \text{eeee}, \text{math}, \text{wyzx}, \dots\}$. Define a function d which takes in two words and outputs the number of positions at which they differ. So for example, $d(\text{math}, \text{cats}) = 2$ and $d(\text{abcd}, \text{dcba}) = 4$.

- (a) Show that (S, d) is a metric space (this is called the *Hamming metric*).
- (b) In (S, d) , consider the sets

$$B_2(\text{aaaa}) = \{s \in S : d(s, \text{aaaa}) < 2\} \quad \text{and}$$

$$D_2(\text{aaaa}) = \{s \in S : d(s, \text{aaaa}) \leq 2\}.$$

Is it true that the closure of $B_2(\text{aaaa})$ equals $D_2(\text{aaaa})$? Using the results from Problem 2 will make things easier.

3.1 Solution (a)

A metric space consists of two parts: a set and a distance function. The metric space is S . The function is d . We need to prove that the distance function satisfies three conditions:

- (a) $d(p, q) > 0$ if $p \neq q$; $d(p, q) = 0$ if $p = q$
- (b) $d(p, q) = d(q, p)$
- (c) $d(p, q) \leq d(p, r) + d(r, q)$, for any r in S

Start with the first.

Proof. We can view words as a finite sequences of letters ($p = \{p_1, p_2, p_3, p_4\}$ and $q = \{q_1, q_2, q_3, q_4\}$). We can view the distance function as an alteration of the discrete function (f).

$$f : X \rightarrow \{0, 1\} \tag{10}$$

$$f(a, b) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases} \tag{11}$$

$$d(p, q) = \sum_{i=1}^4 f(p_i, q_i) \tag{12}$$

Suppose that $p \neq q$. Therefore, they differ in at least one position. Since they differ in at least one position, the distance function would be greater than or equal to 1. Therefore, it would be greater than zero.

Suppose $p = q$. There would be no positions in which the two differ.

$$d(p, q) = 0 + 0 + 0 + 0 = 0 \tag{13}$$

Therefore, the first is satisfied.

OEΔ

Then the second.

Proof.

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4 Problem 4

Let $E = \mathbb{Q} \cap [0, 1]$, viewed as a subset of $(\mathbb{R}, d)$, where d is the usual metric.

- (a) Find an open cover of E that has no finite subcover, and conclude that E is not compact. (*Hint:* pick an irrational number $r \in [0, 1]$ to help you.)
- (b) Find a set $A \subset E$ (other than $\emptyset$ or E) that is clopen w.r.t. the subspace metric on (E, d) , and conclude that E is not connected. You may again find picking an irrational $r \in [0, 1]$ helpful.
- (c) With respect to $(\mathbb{R}, d)$, is the set A you chose in part (b) open, closed, both, or neither? Explain.

5 Problem 5

Let (X, d) be an arbitrary metric space.

- (a) Show that a finite union of compact sets in (X, d) is compact. Is this also true for arbitrary unions? Prove or give a counterexample.
- (b) Show that a set $K \subset (X, d)$ is compact if and only if every cover of K *consisting of open balls* has a finite subcover. (*Hint:* For the trickier direction, given an open cover $\{U_\alpha\}_{\alpha \in A}$ of K , make a new cover that consists of all of the open balls contained in some U_α .)
- (c) Show that compact $K \subset (X, d)$ have the following property: for all $\epsilon > 0$, there is a finite set $\{x_1, \dots, x_n\} \subset K$ so that each $y \in K$ is within distance ϵ of some x_j .

5.1 Solution (a)